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Dear Editors of the SIAM Journal on Mathematics of Data Science,

Please consider the manuscript “Hierarchical Convexity-Run Decomposition of Sampled Curves: Finite-Sample Recovery and Stability.” The paper introduces a nonlinear multiscale representation for sampled curves based on runs of discrete curvature sign. Its main contributions are an exactly reconstructing nested transform with a sparse linear-work implementation, a precise characterization of the discontinuity of hard knot decisions together with stable companion summaries, and finite-sample recovery guarantees for the hierarchy and for an explicit alternating chord-lobe model.

I believe the manuscript is particularly well suited to SIMODS because its central question lies in mathematical data science: how to construct, analyze, and statistically certify an interpretable representation whose atoms are localized convex and concave departures from chords. The theory is complemented by simulations that test the recovery boundary and by a no-refit transfer study on expert-labelled chromatographic curves. That application serves as a downstream validation setting for a general sampled-curve representation, with matched controls and negative boundary experiments reported explicitly.

Code, protocols, compact results, and reproducibility instructions are archived on Zenodo at [doi:10.5281/zenodo.22171977](https://doi.org/10.5281/zenodo.22171977); active development is available at github.com/Safreliy/hcrd. The manuscript is original and is not under consideration elsewhere.

Sincerely,

Savelyi Baturin
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